WEEK 10 – Power BI

Real Estate Analytics Dashboard

King County House Sales Analysis

- **APARNA F**
B.Sc Data Science & AI

Introduction

In Week 10, the Real-Estate Analytics Dashboard was developed using Power BI as a continuation of the Week 9 Housing Data Cleaning and Transformation project. The cleaned King County House Sales Dataset was used to analyse house prices, property characteristics, geographical locations, and sales patterns. DAX measures, interactive charts, map visualizations, and dynamic parameter sliders were used to create an informative dashboard. The dashboard helps users understand different factors affecting house prices and compare property sales across different categories.

Problem Statement

Real-estate managers want to understand property prices, locations, house characteristics, and sales patterns using an interactive Power BI dashboard. The housing dataset is analysed to identify important factors that influence house prices and support better understanding of the real-estate market.

Objectives

- Create DAX measures to calculate important housing indicators.
- Analyse total properties sold, average house price, and median price.
- Calculate average living area and average price per square foot.
- Create geographic map visualizations using latitude and longitude.
- Analyse house prices based on property grade, condition, and waterfront availability.
- Create price distribution charts based on bedrooms and bathrooms.
- Implement dynamic parameter sliders for square-footage filtering.
- Develop an interactive House Sales Analytics Dashboard.

DAX Measures Creation

DAX measures were created in Power BI to calculate important housing statistics. These measures help summarize the housing data and provide useful insights into property sales.

The following measures were created:

Total Properties Sold – DAX Measure

A DAX measure named Total Properties Sold was created using the DISTINCTCOUNT function in Power BI. This measure calculates the total number of unique properties using the property ID column. It avoids counting repeated property IDs more than once and provides an accurate count of unique properties. The measure is displayed in a KPI card to give a quick overview of the total properties in the housing dataset.

DAX Formula :

```
Total Properties Sold = DISTINCTCOUNT('kc_house_data'[id])
```

Average House Price – DAX Measure

A DAX measure named Avg House Price was created using the `AVERAGE` function in Power BI. This measure calculates the average selling price of all properties using the Price column. It helps understand the average value of houses in the dataset. The measure is displayed in a KPI card to provide a quick overview of the average house price.

DAX Formula:

```
Avg House Price = AVERAGE('kc_house_data'[price])
```

Average Living Area – DAX Measure

A DAX measure named Avg Living Area was created using the `AVERAGE` function in Power BI. This measure calculates the average living area of all properties using the `sqft_living` column. It helps understand the average size of houses in the dataset. The measure is displayed in a KPI card to provide a quick overview of the average living area of the properties.

DAX Formula:

```
Avg Living Area = AVERAGE('kc_house_data'[sqft_living])
```

Average Price per SqFt – DAX Measure

A DAX measure named Avg Price per SqFt was created using the `AVERAGEX`, `FILTER`, and `DIVIDE` functions in Power BI. This measure calculates the average price per square foot for all properties. The `FILTER` function considers only properties with a living area greater than zero, while `DIVIDE` calculates the price per square foot by dividing the property price by its living area. This measure helps compare property values based on their living area and is displayed in a KPI card.

DAX Formula:

```
Avg Price per SqFt =
AVERAGEX(
   FILTER(
      'kc_house_data',
      'kc_house_data'[sqft_living] > 0
   ),
   DIVIDE(
      'kc_house_data'[price],
      'kc_house_data'[sqft_living]
   )
)
```

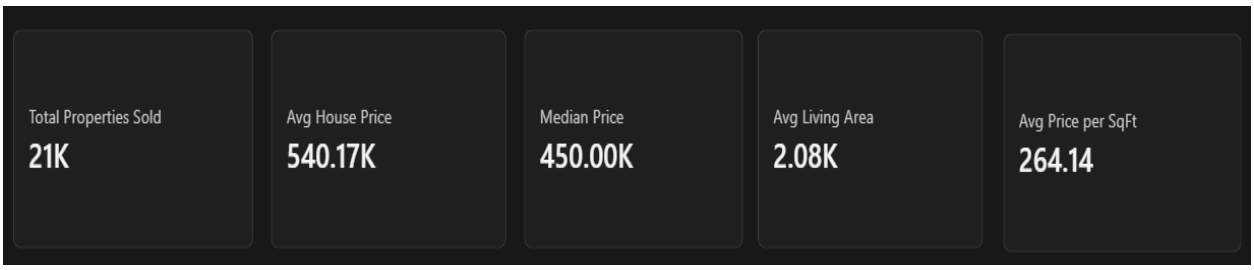
Median Price – DAX Measure

A DAX measure named Median Price was created using the `MEDIAN` function in Power BI. This measure calculates the middle value of all property prices in the dataset. It helps understand the typical house price without being heavily affected by extremely high or low property prices. The measure is displayed in a KPI card to provide a quick overview of the median house price.

DAX Formula:

```
Median Price = MEDIAN('kc_house_data'[price])
```

KPI CARDS



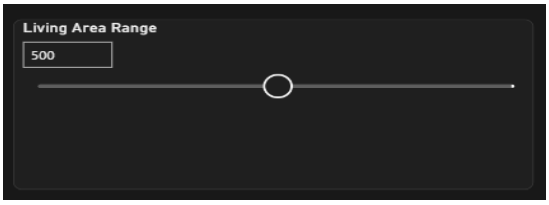
Shape Map – Property Distribution by Price Tier

A Shape Map visual was created in Power BI to represent property categories based on their price tiers. Four categories—Affordable, Budget, Luxury, and Premium—were displayed using different colours. This visualization makes it easier to identify and compare the different price categories on a map.



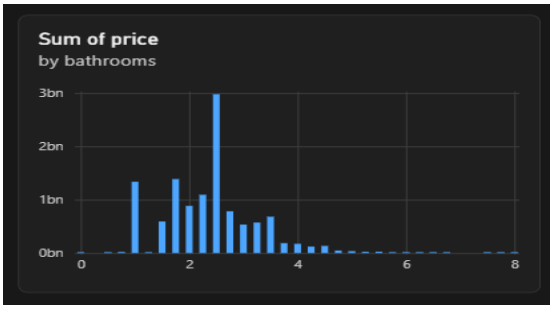
Living Area Range Slider

A numeric range slider was added to the dashboard to filter properties based on their living area in square feet. It allows users to adjust the range and view properties according to their preferred house size, making the dashboard more interactive and useful for analysis.



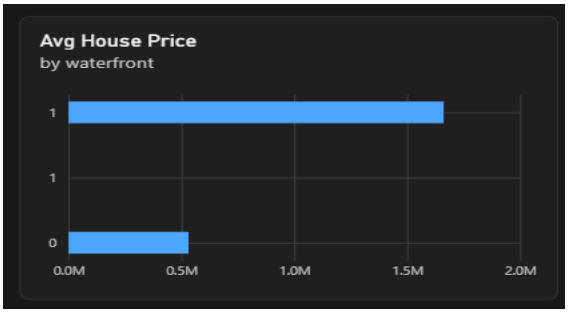
Clustered Column Chart – Sum of Price by Bathrooms

A Clustered Column Chart was created in Power BI to display the sum of house prices based on the number of bathrooms. The vertical columns help compare the total property prices across different bathroom counts. This visualization makes it easier to identify how house prices are distributed among properties with different numbers of bathrooms.



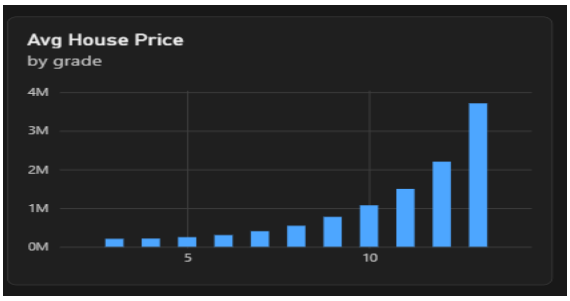
Clustered Bar Chart – Average House Price by Waterfront

A Clustered Bar Chart was created in Power BI to compare the average house prices of properties with and without waterfront views. The horizontal bars make it easy to compare both categories and understand how waterfront availability affects property prices.



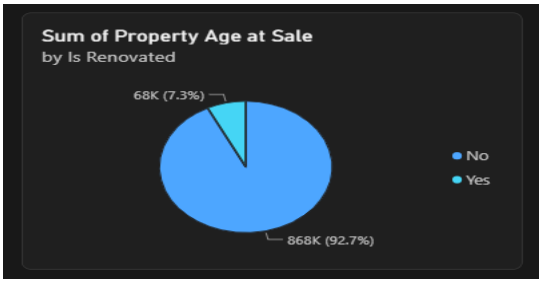
Clustered Column Chart – Average House Price by Grade

A Clustered Column Chart was created in Power BI to display the average house price across different property grades. The vertical bars make it easy to compare prices between grade levels and understand how property quality is related to house prices.



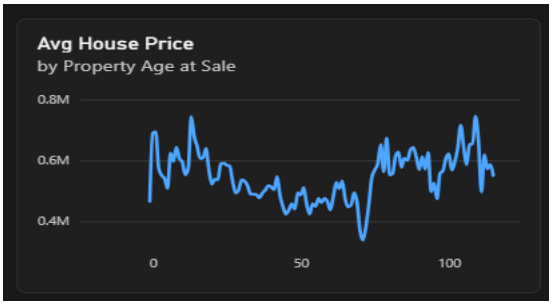
Pie Chart – Property Age at Sale by Renovation Status

A Pie Chart was created in Power BI to show the distribution of renovated and non-renovated properties. The chart shows that 92.7% of properties are not renovated, while 7.3% are renovated. Different colours are used to distinguish the two categories and compare their proportions easily.



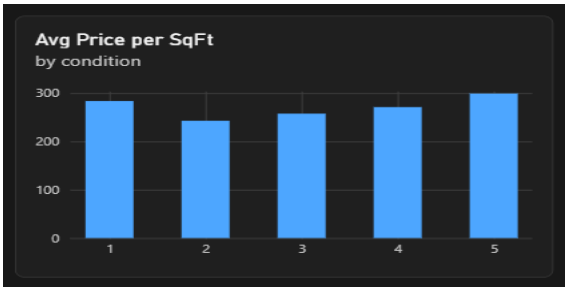
Line Chart – Average House Price by Property Age at Sale

A line chart was created to show the relationship between property age at the time of sale and the average house price. It helps identify how house prices change across different property ages and makes it easier to observe price trends and variations.



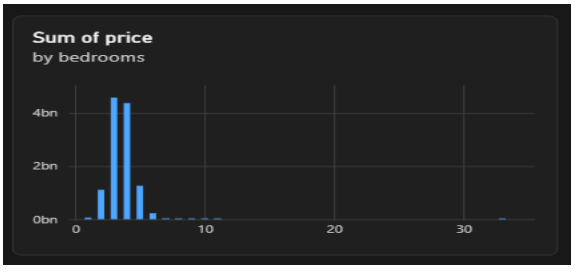
Avg Price per SqFt by Condition – Clustered Column Chart

A clustered column chart was created to compare the average price per square foot of houses across different property condition levels. The chart shows that properties with condition level 5 have the highest average price per square foot, while condition level 2 has the lowest. This helps understand how property condition affects the price per square foot.



Sum of Price by Bedrooms – Clustered Column Chart

A clustered column chart was created to compare the total house prices across different bedroom counts. The chart shows that properties with 2 and 3 bedrooms contribute a large share of the total sales value. It helps identify which bedroom categories have higher overall property sales values and understand the relationship between the number of bedrooms and house prices.

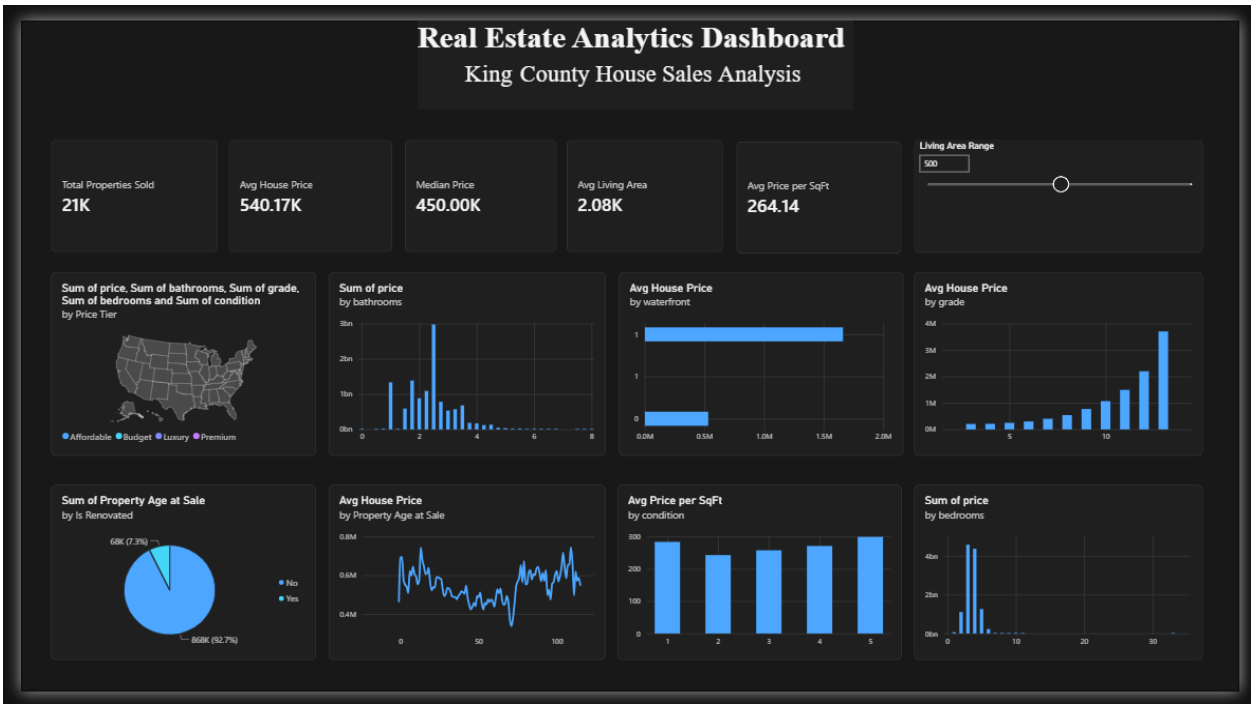


Final Real Estate Analytics Dashboard

The final dashboard was created using Power BI to analyze the King County House Sales dataset. It includes KPI cards showing Total Properties Sold, Average House Price, Median Price, Average Living Area, and Average Price per SqFt. A Living Area Range slider allows users to filter properties dynamically.

The dashboard also contains a Shape Map displaying price tiers, charts comparing house prices by bathrooms, waterfront status, and grade, and visualizations showing renovation status, property age, price per square foot by condition, and total price by bedrooms. These interactive visuals help users understand property characteristics, compare house prices, and identify sales patterns.

Overall, the dashboard presents the housing data in a clear and interactive format, making it easier to explore and understand real-estate trends.



Conclusion

The Week 10 Real Estate Analytics Dashboard was successfully created using Power BI with the preprocessed King County House Sales dataset from Week 9. DAX measures were used to calculate important details such as total properties sold, average house price, median price, average living area, and average price per square foot. Different charts, a Shape Map, KPI cards, and a Living Area Range slider were added to make the dashboard interactive. This project helped in understanding property prices, house characteristics, renovation status, and sales patterns through clear visualizations.